Risk Register

Risk Register — Jeopardy Wagering Model

Last updated: 2026-07-23

Note (2026-07-23): risks #9 and #10 below document a two-layer defect in the Daily Double engine's statistics — the first fix (risk #9) was itself found wrong on a deeper review (risk #10) and re-fixed. Prior conclusions in this register are not rewritten when superseded — each row keeps its own history so the sequence of what changed and why stays legible. See the changelog and `docs/reviews/statistical-review-2026-07-23-v2.md`.

#	Risk	Likelihood	Impact	Mitigation	Status
1	Historical data licensing restricts what appears in shipped artifacts	High	Medium	Cite Tesauro et al. (2013) published statistics for anything shipped; restrict jwolle1 dataset to internal validation only. See ADR-0001.	Accepted, mitigated
2	Real player confidence is unobservable in historical data, limiting how prescriptive the model can be	High	High	Require a self-reported confidence input (e.g., 1–5 scale mapped to historical row-based accuracy) rather than inferring confidence purely from data	Open — design decision needed
3	Live in-game wagering assistance would violate actual contest rules	N/A (policy)	High	Scope the tool explicitly as pre-game study and post-game analysis only; state this in the README and white paper	Open — needs explicit statement in shipped docs

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4	Pairwise heuristic decomposition for 3-player wagers is an approximation, not a proven joint optimum	Medium	Medium	Validate heuristic layer against the full N-player Monte Carlo simulator before treating it as authoritative	Resolved — Tier 1 (`fj_wager.py`) uses a single largest-threat formula, not pairwise decomposition; see section 3.2 of the research memo
5	Tier 2 (`fj_bestresponse.py`) and the Daily Double engine (`dd_wager.py`, `simulate.py`) use illustrative, uncalibrated opponent-behavior and clue-success constants (see ADR-0002)	High	Medium	Every constant is named and documented as a placeholder; hand-derived tests use explicit fixed models, not the defaults, so they stay valid once real calibration replaces the defaults	Open — needs a calibration pass against Tesauro et al.'s full tables and/or the internal-only jwolle1 dataset
6	Monte Carlo win probabilities in the Daily Double engine carry sampling noise, which can make a worse wager look better by chance at low trial counts	Medium	Low	Implemented common random numbers (paired random draws across candidate wagers within each trial) to remove most of this noise; see comment in `dd_wager.py::analyze`	Mitigated — raise `trials_per_wager` for tighter estimates if still needed

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7	Regular-clue and future-Daily-Double dynamics in `simulate.py` ignore buzzer competition and wrong-answer pass-through	Medium	Medium	Documented as a named simplification in the module docstring; results from multi-clue-ahead or multi-DD-ahead rollouts should be treated as rough, not precise	Open — refine before trusting rollouts far from the last Daily Double

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8	`app/index.html` is a hand-written JavaScript port of the Python engine, not generated from it – the two can silently drift apart as either one changes. The two engines also use different PRNGs (Python: Mersenne Twister; JS: mulberry32), so their Monte Carlo Daily Double outputs are only distributionally comparable, never identical trial-for-trial.	Medium	High	Re-checked 2026-07-23 (v2 review): extracted the `<script>` block and ran it under Node against the FJ hand-derived values, the strict-win tiebreak, the 3-way "both leaders bust" case, the inverse-normal and Bonferroni-band reference values, and the paired-SE <= independent-SE invariant – all matched. Still a manual step, not automated in CI.	Open — add a CI step (once CI exists) that runs the shared cases through the JS engine and fails on mismatch; only the exact/deterministic functions can be checked for equality, MC outputs must be checked distributionally

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9	Daily Double engine picked the "best" wager by comparing noisy Monte Carlo win-probability estimates with a fixed 1e-9 tolerance, treating them as exact. Reported 2026-07-23: a 2-point change in stated confidence (42% → 44%) swung the recommended wager from $350 to $4,200 for identical scores – a real defect, not a data problem.	High (was live in shipped app)	High	First fix (2026-07-23, v1): standard-error-aware "plateau" selection. **Superseded by the v2 fix below — the v1 fix's SE formula was itself wrong (see risk #10). Kept here unrewritten for history.**	Resolved by v1, then found insufficient — see risk #10

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10	The v1 fix for risk #9 computed the wager-comparison standard error with the INDEPENDENT two-sample formula `sqrt(se_w^2 + se_best^2)`, but the simulator uses common random numbers (CRN), under which candidate wagers are strongly positively correlated. The independent formula therefore overstated the comparison variance, producing a spuriously wide plateau and systematic UNDER-betting (it recommended ~$0 even where betting was clearly better; for `[4200,6000,3400]`@0.42 the independent rule recom-	High (was live after the v1 fix)	High	Fixed 2026-07-23 (v2) in `engine/dd_wager.py` and ported to `app/index.html`: the comparison SE is now the PAIRED per-trial difference (candidate vs. empirical best), which respects CRN; and the band is a multiplicity-adjusted Bonferroni value `z* = Phi^{-1}(1 - alpha/k)` over the `k` comparisons-with-the-best (Hsu, 1996). Verified stable across seeds and neighbouring confidences on a flat scenario ($0) AND that it still recommends a positive wager on a clear-signal scenario ($1,200). See `docs/reviews/statistical-review-2026-07-23-v2.md`.	Resolved — Python and JS fixed; 33 tests passing incl. a paired-SE <= independent-SE invariant

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11	The Daily Double engine's reported `win_probability` is the in-sample maximum over candidate wagers, so it is an upward-biased ("winner's curse") estimate of the true best win probability	Medium	Low	Documented in the engine docstring, the returned notes, and the v2 review. The *recommended wager* is well insulated (smallest wager on a multiplicity-controlled plateau, not the raw argmax); only the reported percentage is optimistic.	Accepted, documented — a debiased (data-split) estimate is the future fix
12	Tier 2 Final Jeopardy enumeration treats each opponent's correctness as independent, but real FJ correctness is positively correlated across players (shared clue difficulty)	Medium	Medium	Documented against ADR-0002 and in the v2 review. Positive correlation would raise joint "both miss" probabilities, so figures like the 1/12 "both leaders bust" example are conservative floors, not over-estimates.	Open — add an opponent-correctness correlation parameter and report sensitivity

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13	The Tier 2 enumeration is described as “exact”; it is exact in the sense of no sampling error, but is computed in IEEE-754 double precision, not exact rational arithmetic	Low	Low	Accumulated floating error (~1e-16) is far below the 1e-9 comparison tolerance, so no result is affected; white-paper wording tightened to “exact enumeration in double precision (no sampling error)”.	Accepted, documented